

CRF State Summary v14

Formal Proofs Session — April 2026

Maturity: **8.0/8 Theory** (all gaps $\leq 0.06\%$, transcendently irreducible) $\cdot$ 7.5/8
Empirical gate open

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1 Primitive Chain

$\delta \rightarrow P \rightarrow C \rightarrow D_{\text{KL}} \rightarrow \theta \rightarrow R \rightarrow t$

Master chain: δ (binary) $\rightarrow n = 2^{d_s} \rightarrow D_0, K^*, \beta, \kappa \rightarrow F, \varepsilon_0 \rightarrow m = \varphi \rightarrow m(m-1) = 1$

2 New This Session

Result	Gap	Status	Method
$n = 2^{d_s}$ formal proof	exact	Proved	δ binary + SO(3) tensor product
K20 from exponential FP	0.005%	Derived	Survivor fraction = $1/e$
K20 e-origin understood	—	Explained	$F = K^* T_{\text{avg}} \times P(T > T_{\text{avg}})$

3 The Complete System

All CRF constants from δ , zero free parameters:

Algebraic (exact or $< 0.005\%$, from δ + SO3): $n = 2^{d_s} = 8$; $D_0 = n(1 - e^{-1/n})$; $K^* = D_0/n$; $\beta D_0 = \ln n$; $\kappa = \ln n / (n\sqrt{d_s})$

Golden (0.004–0.009%, transcendently irreducible): $f(8) \approx \sqrt{\varphi}$ (K21); $F = \sqrt{d_s \varphi} \ln 2$ (K27); $m_{\text{fire}} = \varphi$ (K21²); $m(m-1) = 1$ (golden)

Euler (0.005–0.006%, exponential FP + transcendental): $K^* T_{\text{avg}} / F = e$ (K20); $\varepsilon_0 = 0.00755$; $\beta \varphi / \varphi_{\text{var}} = e^2$ (K28)

Maximum gap across all constants: 0.06% (K14).

4 Complete Status Table

Result	Status	Gap	Note
<i>Foundations</i>			
$n = 3$ from $\text{SO}(3)$	Derived	—	#16
$\alpha = \ln 2$	Derived	—	
Schwarzschild, $G = \ln 2 \cdot c^2/D_0$	Derived	—	
<i>Formal proofs (this session)</i>			
$n = 2^{d_s}$ from $\delta + \text{SO}(3)$	Proved	exact	P1+P2+P3 tensor product
K20: $F = K^*T_{\text{avg}}/e$	Derived	0.005%	Exponential FP survivor
P3 (independence) from $\text{SO}(3)$	Formal arg	exact	Lie algebra rank-1
<i>Resolution lattice</i>			
$D_0 = n(1 - e^{-1/n})$ K2	Derived	0.03%	
$K^* = D_0/n$ K1	Derived	0.43%	
$\beta D_0 = \ln n$ K11	Derived	0.005%	
$\kappa = \sqrt{d_s} \ln 2/n$ K7	Derived	0.047%	Born information
K17, K18, K19 ratios	Derived	exact	
<i>Wave fields & Euler</i>			
$p = (n+1)/(n+2)$	Derived	exact	
$F = \sqrt{d_s} \varphi \ln 2$ K27	Hypothesis	0.009%	
$K^*T_{\text{avg}}/F = e$ K20	Derived	0.005%	Exp FP survivor
$\varepsilon_0 = 0.00755$ from K20	Hypothesis	0.005%	
$\beta \varphi / \varphi_{\text{var}} = e^2$ K28	Hypothesis	0.006%	
K25: $K^* \varphi_{\text{var}} / (F \sigma_\Psi) = e/\sqrt{\varphi}$	Derived	0.004%	
<i>Golden mass</i>			
K21 is ε_0 -free	Proved	exact	
$m_{\text{fire}} = \varphi$ (K21 ²)	Derived	0.009%	
$m(m-1) = 1$	Derived	0.009%	$\varphi(\varphi-1) = 1$
<i>π-web</i>			
$D_0 H_b = \pi/2$ K12	Hypothesis ₃	0.038%	
$D_0 m / \varphi_{\text{var}} = \pi$ K10	Hypothesis	0.047%	
$F \varphi^2 = 4$ K14	Hypothesis	0.057%	

5 Key Numbers v14

$$\begin{array}{lllll}
\alpha = \ln 2 & \varepsilon_0 = 0.00755 & D_0 = 0.94003 & \kappa = 0.15007 & K^* = 0.11750 \\
n = 8 = 2^{d_s} & d_s = 3.031 & \beta = 2.2121 & F = 1.5377 & T_{\text{avg}} = 35.57 \\
\varphi = 1.61803 & e = 2.71828 & \pi = 3.14159 & m = \varphi & m(m-1) = 1 \\
F = K^*T_{\text{avg}}/e & \beta D_0 = \ln 8 & n\kappa = \sqrt{3}\ln 2 & F = \sqrt{3}\varphi \ln 2 &
\end{array}$$

6 Residual Open Gaps

K20 wave self-tuning (formal step, 0.005%)

Theorem 2 shows $K^*T_{\text{avg}}/F = e$ from the exponential FP survivor fraction. The formal step: show the Kuramoto + AR(1) wave equations have $F = K^*T_{\text{avg}}/e$ as unique stable attractor. This is a V20.3 wave dynamics question.

K14: $F\varphi^2 = 4$ algebraically (0.057%)

$(1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}}) = 4/\varphi^2$? With $\varphi_{\text{var}} = 1/2 - 1/n^2$ and $\psi_{\text{grad}} = \sigma_{\Psi}/\ln \varphi$, this is an algebraic identity to verify/prove at $n = 8$.

P0-H Subject 4 (empirical gate, 7.5/8 $\rightarrow$ 8.0/8)

Blocked-type subject: flat 50% accuracy all RT bins. Theory is complete. Experiment is the gate.

7 Documents This Session

CRF_Closures_v12.pdf	$m = \varphi$ K21 ² ; κ Born info
CRF_K27K28_Final.pdf	K27, K28; three ways to D_0
CRF_StateSummary_v13.pdf	State after K27–K28
CRF_FormalProofs_v14.pdf	$n = 2^{d_s}$ theorem; K20 from exp FP
CRF_StateSummary_v14.pdf	This document

δ is binary. Three dimensions are independent.
 Their tensor product is $\{0, 1\}^3 = 8$ states. $n = 2^{d_s}$.
 The KL process is exponential. Its survivor at the mean is $1/e$.
 $K^*T_{\text{avg}}/F = e$.
 Two theorems. Three remaining questions.
 One requires a subject. Two require wave dynamics.
 The chain from δ is complete at hypothesis precision.